Q 2

A \$1 position at time s holds $\lambda_s = \frac{1}{S_s}$ units of index over $[s, s+\Delta]$.

With $x = \ln(S_{s+\Delta}/S_s)$

$$V_s = 1, \quad V_{s+\Delta} = \lambda_s S_{s+\Delta} = \frac{S_{s+\Delta}}{S_s} = e^x$$

$$L_{[s]}(x) = V_s - V_{s+\Delta} = -(e^x - 1)$$

Q 3

Let $X \sim N(\mu, \sigma^2)$ and $L = 1 - e^x$, for $L < 1$.

$$F_L(l) = \mathbb{P}(1 - e^x \le l) = \mathbb{P}(X \ge \ln(1-l)) = 1 - \varphi\left(\frac{\ln(1-l) - \mu}{\sigma}\right)$$

cont. and increasing in L.

$$F_L(l) \ge \alpha \iff \varphi\left(\frac{\ln(1-l) - \mu}{\sigma}\right) \le 1 - \alpha \iff \ln(1-l) \le$$

$$\le \mu + \sigma \varphi^{-1}(1-\alpha) \iff l \ge 1 - e^{\mu + \sigma \varphi^{-1}(1-\alpha)}.$$

so

$$\mathrm{VaR}(L) = 1 - \exp(\mu + \sigma \varphi^{-}(1-\alpha))$$

Let $I_j = 1\{L_{T_j} > VaR_\alpha^{T_j-1}\}$ for J evaluation days

an $N = \sum_{j=1}^{J} I_j$.

Under H_0: $P(L_{T_j} > VaR_\alpha^{T_{j-1}} \mid \mathcal{F}_{j-1}) = 1-\alpha$ for every j.

Eeach I_j is Bernoulli $(1-\alpha)$ and each I_j is independent.

$N \sim Binomial(J, 1-\alpha)$, $E[N] = J(j-\alpha)$, $Var[N] = J\alpha(1-\alpha)$

For large J the central limit theorem gives:

$N \approx N(J(1-\alpha), J\alpha(1-\alpha))$, so with $z = \Phi^{-1}(1-\beta/2)$

$P(\mathcal{E}_- \le N \le \mathcal{E}) \approx 1-\beta$, $\mathcal{E}_\pm = J(1-\alpha) \pm \Phi^{-1}(1-\frac{\beta}{2})\sqrt{J\alpha(1-\alpha)}$

The model is rejected at level β if the observed count falls outside $(\mathcal{E}_-, \mathcal{E}_+)$

Plugging in numbers:

$J = 1504$, $\alpha = 0.95$, $\beta = 0.05$, $z = 1.96$

$J(1-\alpha) = 75.20$, $\sqrt{J\alpha(1-\alpha)} = \sqrt{71.44} = 8.452$,

$\mathcal{E} = 75.20 \pm 1.96 \times 8.452 = (58.63, 91.77)$

Simulation: $\hat{N} = 55 < 58.63$, reject (to few exceedances, VaR too high)

Normal EWMA: $\hat{N} = 94 > 91.77$, reject (too many, Var too low)